

**Two Innovation Systems, One AI Rivalry: Comparing Artificial Intelligence Development
and Its Economic and Political Consequences in China and the United States**

Ziheng Wang (William)

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Dr. Hartlep

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Abstract

Public discussion often regards the rivalry between the United States and China in artificial intelligence (AI) development as a simple technological race, overlooking the interactions between different organizations and governmental institutes. This paper aims to analyze this competition, its economic and political implications more comprehensively by adopting the Comparative National Innovation System framework. While both nations share common pressures, including GPU hardware bottlenecks, soaring electricity demand, and talent competition, their pathways are also divergent in research, funding, infrastructure, capitals, and politics. This is a result from their distinct institutional architectures reflected via NIS: China's mechanism is state-coordinated and power-centralized, while the U.S. side is mainly market-driven and decentralized. These logics generate varied consequences: in the United States, heavy macroeconomic reliance on AI raises risks while data center demand drives up electricity prices; in China, resource misallocation and underutilized facilities impose fiscal burdens on local governments. Internationally, both nations are attempting to diffuse their models abroad as a geopolitical strategy.

Keywords: Artificial Intelligence, China, United States, National Innovation Systems, AI Governance, Frontier Models

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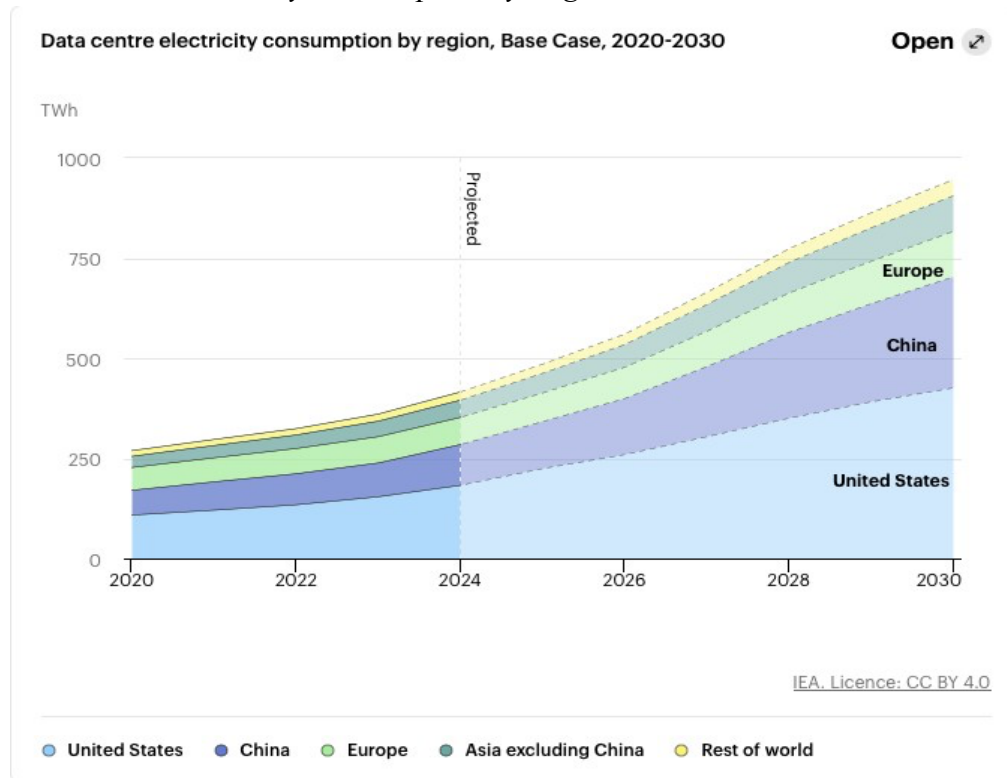
Background Knowledge

As humanity steps into the age of digital society, Artificial Intelligence (AI), referred to as “technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy” (IBM, 2026, para. 1), has been widely adopted into various industries and routine life to promote productivity and redirect people’s energy into more valuable tasks. Realizing AI’s strategic significance, countries across the globe have been investing substantial amounts of financial and manual resources in related fields. AI infrastructure, such as “data centers” and “electric grids” are continuously being constructed. Data centers house servers which can store digital workloads and process data via high-performance Graphics Processing Units (GPUs) (International Energy Agency [IEA], 2025a). Electric grids provide sufficient energy that enables data centers for AI conducting such actions. Additionally, the study towards semiconductors and chips has been a heated topic, as such hardware directly determines the maximum compute power and performance an AI could possibly manifest.

Figure 1 below illustrates that the United States and China will remain as the two most energy-demanding countries in the next few years, taking up 45% and 25% of the global consumption, respectively. Undoubtedly, the AI developments of these two nations are the most discussed in news reports and scholarly research.

Figure 1

Data Center Electricity Consumption by Region, Base Case, 2020–2030



Note. Adapted from *Data Centre Electricity Consumption by Region, Base Case, 2020–2030*, by International Energy Agency (IEA, 2025a).

Each has different economic and political advantages and weaknesses that could facilitate or hinder the development of this powerful tool. Governments and institutions have been striving in the past few years, trying to take the lead and outcompete their counterparts. Policies regarding AI development and regulations, such as “American’s AI Action Plan” and “AI Plus” Initiative in China, have been made and partly fulfilled in both nations, encouraging companies to continue creating more powerful Large Language Models (LLM), which is defined as “a statistical model of a large training corpus that creates new content from prompts by generalizing past examples” (Shapiro, 2025, p. 2).

Research Problem and Purpose

Discussions being had by the general public frequently treat AI development as a simple technological race while underexamining the national systems through which research, capital, infrastructure, regulation, labor, and state power are organized. This facile understanding is a problem because it misunderstands how accurate comparisons of China and the United States are made. Consequently, the purpose of the paper is to compare the United States and China's national innovation systems, specifically, how they affect technological development, and trace the economic and political consequences emerging from their similarities and differences. This research, which applies Freeman's (1987) National Innovation System (NIS) framework to study AI in the United States and China, is beneficial and contributes to the current literature base because it considers the interactions between different actors and avoids oversimplifying the effect of certain policies.

Research Question

This paper aims to answer the question, "How do similarities and differences in the national innovation systems of the United States and China shape their respective approaches to artificial-intelligence development, and what economic and geopolitical consequences are emerging from those differences?"

The United States and China have completely different social circumstances and political and economic systems, inducing distinctive preference and directions for AI development. The Chinese government has enacted compulsory regulations and allocated resources through state-coordinated logics, while its U.S. counterpart has tended to hand most of the work to the capitalistic market and manipulated certain sides via an indirect approach. The two mechanisms will eventually lead to varied distribution of benefits and harms to social groups.

Organization of the Paper

This paper follows a standard research report format. First, the research method and data sources are shared. Then, the author introduces the theoretical framework used in this paper, which is the National Innovation System (NIS) (Freeman, 1987). Next is the literature review, which shares a high-level summary and syntheses of research papers, news reports and governmental files. Next, the paper shares a comparative analysis between the United States and China utilizing NIS. This is followed by sharing economic and political implications. Lastly, the limitations of the research are discussed as well as considerations for future research.

Method and Data Sources

Comparative Research Design

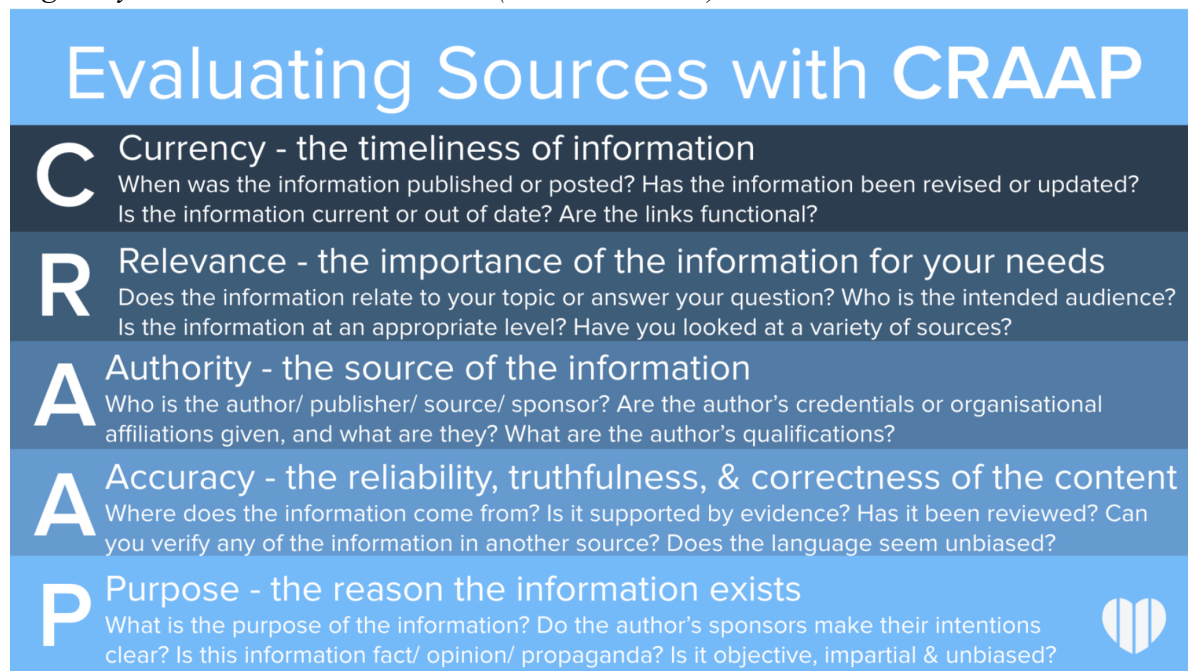
To answer this paper's research question, an NIS framework is utilized in order to compare and contrast these two countries' national innovation systems related to AI. There are six dimensions to Freeman's (1987) NIS framework, which are elaborated on and analyzed within this research paper, covering aspects including but not limited to the following: financial investment, regulatory policies, and technology industry. These dimensions can fully illustrate the interactions and interdependency between actors. (Stanford Institute for Human-Centered Artificial Intelligence [Stanford HAI], 2025).

Data Sources

The author obtained varied types of sources from Google Scholar, JSTOR, Oberlin Library, and News Agency by searching key terms including "national innovation system," "artificial intelligence," "China AI policy," "United States AI investment," "AI data centers," "semiconductor export controls," "AI governance." etc. The time scope of this search was limited between 2017 and 2026, covering most sources published in the fields of Artificial

Intelligence. All information needs to be verified as credible before adopting into the paper using the CRAAP framework (Blakeslee, 2004) and source engagement methodologies outlined by Booth et al. (2024). All references shall be vetted for authority, empirical accuracy, currency, and alignment with the analytical dimensions of the National Innovation Systems framework. The inclusion criteria is categorized and described in detail in Figure 2:

Figure 2.
Eligibility Criteria Drawn on CRAAP (Blakeslee, 2004)



Note. Adapted from *Cite This For Me, The CRAAP Test: An Easy & Fun Way to Evaluate Research Sources*

After the eligibility verification process, references are recorded and categorized into different types. As table 1 demonstrates, the author analyzed 116 references in total (not concerning 2 AI usage disclosure) from JSTOR, Oberlin library, and Google Scholar. These references can be categorized into the following types/subtypes: (a) primary policy/legal documents, such as America's AI Action Plan in 2025 ($n = 11$); peer-reviewed academic and research papers ($n = 16$); institutional reports like Stanford 2025 HAI Index and government statistics ($n = 36$); consulting forecasts from renowned companies including Goldman

(via CNBC) and Morgan Stanley ($n = 5$); journalism / authorized analyst blogs ($n = 48$). The data can also be categorized into what the research has categorized as Foundational Theory (FT) or Empirical Evidence (EE). FT refers to the theoretical concept of the National Innovation System in the United States and China, supporting the general operation of frameworks, while EE refers to the case studies and evidence of such systems in reality, providing sufficient amounts of empirical data supporting the claims. FT is concentrated in the peer-reviewed academic research articles, and EE is useful in all types of sources.

Table 1.
Sources of Data

Reference Type	Number Analyzed (n)	Example of Publication	Empirical Evidence (EE) or Foundational Theory (FT)
Policy/Legal Document	11	America's AI Action Plan in 2025 (The White House, 2025)	11 = EE 0 = FT
Peer-Reviewed Academic Research Articles	16	National Systems of Innovation: Towards a Theory of Innovation and Interactive Learning (Lundvall, 1992)	7 = EE 9 = FT
Journalism, Blogs	48	DeepSeek unveils V4 model, with rock-bottom prices and close integration with Huawei's chips (Fortune, 2026)	48 = EE 0 = FT
Government statistics / Institutional Reports	36	Artificial Intelligence Index Report 2026 (Stanford HAI, 2026)	35 = EE 1 = FT
Consulting Forecasts	5	U.S. data center power demand projected to double by 2027 (Goldman Sachs, 2026)	5 = EE 0 = FT

The data were organized using the six dimensions shared within the National Innovation System work carried out by Freeman (1987), which are as follows: (1) technical side, (2) infrastructure and data, (3) human capital, (4) financial capital, (5) governmental policies, (6) interactions between actors,. This was done intentionally a priori in order to structure the literature review and comparative analysis in a logical and organized way.

Analytic Procedure

After determining which part of the research question and the six dimensions a source belonged to, the researcher read the articles and made annotations in Google Docs. Only the original text of arguments, facts, data, and policies were recorded to preserve its originality, along with the summary of key insight for each passage to assist in comprehending the synopsis quickly in later writing process. Claude Opus 5.0 from Anthropic was used during this stage of the analysis process, initially for the purpose of formatting the document and aligning certain detailed information with annotations. Since Claude Opus 5.0 has limitations, such as hallucinating and producing inaccurate output, the researcher inserted himself into what has been called “human in the loop” (Monarch, 2021) to double-check and confirm the accuracy and safeguard the precision of the analysis and write up. In total, 47 references were uploaded to, and run through, Claude Opus 5.0, producing 30 pages of annotations and analytic memos. The author then used these annotations to horizontally compare and contrast certain features of the two innovation systems by categorizing all bibliography under the six analytical dimensions. This method makes the discrepancies and commonalities apparent, facilitating the writing process and comparative analysis.

Comparative National Innovation Systems Framework

Origins and Development of National Innovation Systems

In his book *Technology Policy and Economic Performance: Lessons from Japan*, Freeman (1987) firstly introduced the modern concept of a “National system of innovation,” defined by him then as “the network of public- and private-sector institutions whose interactions initiate, import, modify, and diffuse new technologies” (p. 1). This definition established the claim that innovation is a systemic and nationally organized procedure instead of the final product of isolated stakeholders.

Eight years later, Freeman (1987) traced the concept back to Friedrich List (1841), a German economist who wrote *The National System of Political Economy*. Instead of completely contending with the then-dominant, free-trade theory – wherein markets shall run freely and wealth will follow – List (1841) insisted that lagging countries like Germany who were trying to catch up with Britain, should build up their productive capacity, including but not restraining education, infrastructure, and commercial industries, before it competed with other countries. These earlier ideas and developments of innovation correspond with the foundational premise of the NIS: national institutions interact with each other with the purpose of creating and absorbing new technology.

Building further on NIS, Bengt-Åke Lundvall (1992) shifted the primary focus from structures to the process of institutions named as “learning.” He believed the fundamental activity driving innovation is learning happening between institutions through exchange, giving an example of “user-producer relationships,” wherein a “producer” makes something, and the “user” provides feedback information about the “using experience”— what is good about this product, what could be improved, and what should producers do next? Thus, both sides benefit

from the back-and-forth “learning” and technology eventually advances over cycles. Nelson (1993) validated the NIS by case studying the United States, Japan, and South Korea. One aspect Nelson (1993) put most effort in is the field of research and development, studying national research funding, interplay between educational institutes, government, firms, and labs, and the transformation from scientific knowledge into technology. He also stated that each country has an entirely different NIS shaped by its unique social, cultural, and historical context.

Edquist (1997) compiled and consolidated previous research theories around three components: (1) technologies, (2) institutions, and (3) actors. In the same year, OECD turned NIS into a tool for analyzing policy, marking the first application of once academic theory into a governance instrument. OECD also provided a rationale for potential failure in the framework, that connections between actors are broken and missing. When firms, universities, and government agencies fail to share knowledge and coordinate, the system as a whole underperforms, even if each individual actor is functioning.

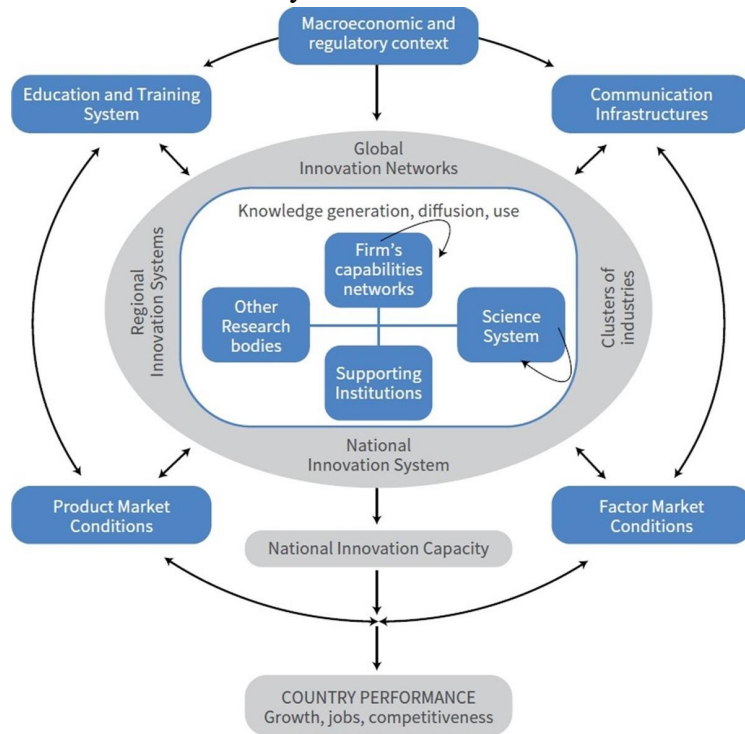
Core Concepts of the National Innovation Systems (NIS) Framework

Based on theories discussed in the previous section, this research operated from the perspective that the National Innovation Systems (NIS) is a framework arranged for the “learning” process to happen between actors while repeatedly interacting with one another to facilitate the diffusion of technology in the broad sense of view.

Figure 3 illustrates a typical system linked for knowledge exchange; the outer grey ring states the scales of activity, regional, national, and international; the peripheral blue boxes being the conditions on which NIS is based, shaped the core and are shaped by it as well; every output produced from this system ultimately channels down into National Innovation Capacity, which is

the overall ability to produce and commercialize new technology in the long run. It is critical to a country's performance, such as economic growth, job markets, international competitiveness, etc.

Figure 3.
National Innovation Systems Framework



Note. Adapted from *Understanding and Comparing National Innovation Systems: The U.S., Korea, China, Japan, and Taiwan*, by R. D. Atkinson, S. Ezell, H. Jeong, Y. J. Kim, and S. B. Kwon, 2025, Information Technology and Innovation Foundation (ITIF).

The following three concepts compose the NIS framework and are thus, important to understand definitionally:

- (1) Technology, which is referred to as the new invention itself;
- (2) Institutions, which are the rules of the system, including laws, social norms and industry standards, providing the incentives and legal parameters in which innovation activity occurs; and

(3) Organizations or actors, which are the actual entities like companies, government agencies, and research labs that communicate with each other. They perform research and development (R&D), commercialization, policy making, and human capital training.

Within the NIS framework, the spread of knowledge undergoes two main phases. Firstly, knowledge emerges within frontier research, lab experiment, and corporation R&D departments. These studies are financially supported by either governmental subsidies or private venture investment. Technological policies might also facilitate this procedure by encouraging the directing of these research developments. Subsequently, knowledge diffuses through organizations and industries within the nation, or even to other countries. Organizations including tech transfer offices, open-source platforms, local governments, labor markets, etc. contribute substantially to this process. Meanwhile, technical standards, licensing policy, national incentives, and industry baselines are the institutions performing as well.

Applying National Innovation Systems to Artificial Intelligence

AI was a heated discussed topic and widely adopted into industry, government bodies, and routine life in the past few years, standing as a perfect example of “technology” in the 21st century, alongside the Internet. Undoubtedly, organizations in different fields are acting to maximize their benefits gained from this tool, while research labs and AI corporations are repeatedly yielding feedback from consumers and analysts to improve their product, illustrating the previously mentioned user-producer relationship. Moreover, governments make policies restricting or promoting the move of research labs and tech companies, while the release of powerful AI models will in turn benefit the application in labs and industry. Such interactions among actors are conspicuous, influential, and indispensable, which fits well into the concept of

NIS. Thus, for reasons stated above, it is entirely feasible and sound to adopt this academic framework in comparative analysis of two countries.

The NIS framework is historically contingent, in List's (1841) words, meaning that it depends on the country's politics, stage of development, past choices, etc. Nelson (1993) agreed by stating that every entity has its distinctive structure of innovation system, rather than a universally correct form. Thus, the NIS framework is ideal for studying how elements are configured and arranged differently or identically between the United States and China in the context of AI development, and in what ways these will lead to the disparities in economic and political outcomes.

Operationalizing the National Innovation Systems (NIS) Framework

Primarily inspired by Nelson (1993) and the OECD analysis (1997), this paper analyzes the Chinese and the United States NIS in topics of AI development and impact from six main dimensions: technical advancement and R&D will be compared between two countries; then it will discuss the infrastructure, especially data center and power grid construction; AI expert flow and regulation in both areas; financial asset and investment from either government or venture capital; regulatory or incentive policy made by central and state governments; four types of communication between actors and organizations. These perspectives will comprehensively cover critical components in the NIS framework, compare and contrast two nations effectively, and help readers comprehend the findings thoroughly.

Differences and similarities both exist in two countries' NIS framework, shaped by their distinct culture, economic & political background, historical development, and technological standards. This will eventually lead to interesting social outcomes, in both positive and negative ways, which requires to be maintained or addressed by the cooperation of actors.

Limitations of National Innovation System (NIS) Framework

Despite the advantages and functions mentioned in previous passages, the NIS framework is sometimes criticized by some analysts. Firstly, the early concept of this mechanism has been described as too vague, open to misinterpretation, and too inclusive to the point of being impractical (Watkins et al., 2015), which can directly undermine the operationality and practicability in actual scenarios. Secondly, the foundation of NIS is mainly based on national scale. However, as globalization and interactions between international entities become increasingly critical in the diffusion of innovative technology, this framework might fail to offer sufficient insight into these larger-scale exchanges. This paper acknowledges that limitation, but concentrates on comparing how domestic actors organize AI development within the United States and China, while cross-border interactions such as international relations and geopolitics, will be addressed with less emphasis.

Literature Review

China's National Innovation System and AI Development

China has a state-coordinated and centralized catch-up system (Liu & White, 2001; Lundvall & Rikap, 2022). The central government in Beijing holds the supreme power of decision making, forming AI policies in a strong mission-oriented way, to offer general developmental guidelines for different states (Gu & Lundvall, 2006). Local governments will then follow the will of central policies to conduct self-driven experimentation of technology, based on the city's or state's economic and social circumstances (Ministry of Science and Technology, 2019; Bird & Bird, 2024). For instance, Shanghai, acting as the most critical financial hub in Yangtze River Delta, emphasized lawful data-trading venues and transformation of finance in 2022 (Standing Committee of the Shanghai Municipal People's Congress, 2022).

Meanwhile, Shenzhen as a Special Economic Zone with a strong hardware and entrepreneurship base, allowed low-risk AI products to enter pilots as soon as the international regulatory standard was met (Standing Committee of the Shenzhen Municipal People's Congress, 2022).

The main actors in the Chinese NIS are party-state agencies, state-owned-enterprise alongside private companies, and public universities (Lundvall & Rikap, 2022). Financial funding is mostly government-led, allowing transparent and explicit steering (Chan et al., 2025). However, this feature makes the state bear most investment risk instead of private investors, since they front the money before the viability is proven, and the price of failure will be absorbed by public balance sheets including fiscal budgets, state banks, and local government (Mazzucato, 2013).

As the backbones of researching and developing frontier models, tech firms have been allocating tremendous amounts of funds into AI development: In 2025, Alibaba, the company currently developing a LLM named Qwen, announced it has put more than 380 billion RMB (around \$53 billion) over three years into the construction of cloud and AI infrastructure (Alibaba Group, 2025). Due to the lack of AI chips caused by U.S. export controls, they need to maximize the efficiency of LLM to make them less compute-demanding, but still able to produce incredible results (Bureau of Industry and Security, 2022; Chan et al., 2025). In order to increase their competitiveness and reputation in the international market, they also make their product open-sourced, which will attract global developers to build their projects based on Chinese models, gradually making it the default, and also allows for rapid downstream application into other industries (Luong, 2026).

Infrastructure in China is well-developed: the national electric grid has been expanding since 2009, providing sufficient energy to AI-associated facilities, even facing surpluses most

times (Sixth Tone, 2026; IEEE Communications Society, 2026). Interactions and cooperations between states are especially emphasized to promote the construction: resource-abundant regions, such as Shanghai, Beijing, and Shenzhen, to assist underdeveloped provinces like Xinjiang and Gansu to construct data centers for AI usage, namely the famous “East Data, West Computing” project (IEEE Communications Society, 2026).

The United States’ National Innovation System and AI Development

On the other hand, as a quintessential capitalist economy, the United States has a decentralized, market-led NIS, which means the White House normally does not control each state and tech companies, but offers voluntary and suggested guidelines instead. The country uses “the invisible hand” – individuals acting in their own self-interest within a free market accidentally promote the overall economic well-being of society (Smith, 1776) – to let the market navigate itself and only intervene during economic failure or initial regulations. In contrast with China having a unified plan, the United States has no single coordinating center in most cases, and states often legislate independently. Some of their regulations and navigated goals might even contradict each other, based on each state’s political parties and economic conditions. California’s Transparency in Frontier AI act (SB 53, 2025), regulates advanced model developers at the training-compute step, imposing risk-reporting duties and penalties up to \$1 million, while the Responsible AI Governance Act (HB 149, 2025) takes an intent-based approach, targeting AI designed for harmful purposes. This could hinder the progress of cooperation between different regions and business between states, forcing companies to comply with conflicting baselines and requirements.

In 2025, data centers will take up 50% of all U.S. electricity demand growth, and the DOE now projects the total U.S. energy demand to rise around 15-20% over the next 10 years.

Thus, the United States has been constructing data centers and expanding the power grid over the past few years to meet rising compute and electricity demand. According to the study of S&P Global, \$47 Billion dollars was invested into the construction process in the past year. On the other hand, China's expansion plan commenced far earlier than the United States, barely facing shortages and rising costs.

Most frontier labs – such as Open AI, Anthropic, and Google DeepMind – are private instead of controlled by the state. Investments from venture capitals (VC) and certain tech companies are the main channel of supplying incoming funds. From 2025 to 2026, OpenAI received \$30B from SoftBank, \$30B from NVIDIA, \$50B from Amazon, and significant financial support from MGX (sovereign capital), TPG (private equity), etc. Uncertainty risks will be borne by those actors instead of federal agencies. The four largest U.S. tech firms, Amazon, Alphabet, Meta, and Microsoft, plan to spend approximately \$725 billion on capital expenditure in 2026, which rose 77% compared to \$410 billion in 2025. Most of the money goes to AI data centers, GPUs, custom silicon, and power.

Shared Pressures and Divergent Pathways

The United States and China have encountered similar challenges and pressures over the past few years, typically shortage of facilities and demand outstripping supply. In terms of hardware, chips specially designed for AI usage are consumed rapidly in both nations, distributed to most demand-intensive data centers (Lundvall & Rikap, 2022). However, the supply for those chips from NVIDIA or TSMC are fixed and fail to meet the rising demand. For China, U.S. export controls blocked advanced chips from entering, enabled it to take the lead in designing compute-intensive models (Bureau of Industry and Security, 2022), while Chinese

tech companies respond with efficiency-optimized models, domestic substitutes made by Huawei and SMIC, and architectural workarounds (Chan et al., 2025).

Secondly, electric energy is another restraining factor for AI development. Power demand has been exploding in both regions as data centers are consuming tremendous amounts of energy (Goldman Sachs, 2026). China has been expanding its national power grid in the past few decades, so it sometimes faces surplus in energy and overcapacity, while some data centers remain underutilized, indicative of misallocation of resources (Australian Strategic Policy Institute, 2026). On the other hand, the United States is currently building up power after realizing its importance before long, with shortages already severe in some demand-intensive regions (Institute for Energy Economics and Financial Analysis, 2025).

Moreover, both nations are facing shortage in AI professionals, competing for the same global pool. The United States attracts them via universities and immigration, and the fact that it's stronger at frontier research, while the eastern counterpart builds their talents domestically, with some proportion of returnees from other countries, but still losing a large amount of STEM elites (Lundvall & Rikap, 2022).

Public legitimacy and trust, including safety, job markets, and surveillance over AI models, need to be sustained. The United States uses a contested bottom-up control: making rights-based rules, litigation, public debate, and fragmented state laws (Baker Botts, 2026). In contrast, China sustains legitimacy via central-state-coordinated governance, content control, and stability-oriented regulations, which are typical of top-down regulations (Papyshev & Yarime, 2023).

From the comparisons above, it is apparent that both nations have common difficulties while making progress in AI development, but they adopted entirely different approaches to

resolving those issues, mostly based on their unique economic and political circumstances. More detailed analysis and examples will be presented in the next section.

Gap in the Literature and Contribution of This Paper

Despite most sources effectively managing to illustrate actions of different organizations and rules of institutions in the United States and China, they are prone to oversimplify the implications or causes of these actions, neglecting that the entire NIS is an interconnected network, not merely actions conducted by one actor (i.e. governmental policies, tech companies model announcement). Consequently, this paper aims to synthesize sources from all industries and perspectives to provide insight into the full operational system, instead of focusing on micro-level changes. Dimensions of comparative analysis are the six layers to observe and analyze this topic comprehensively, demonstrating how one aspect is, in fact, closely related to every other action, and how a decision made by one actor will be restrained by yet also affect the entire industry, economic conditions, and policymaking.

Comparative Analysis and Findings

Technical Design & Research and Development

The objectives of AI development in China and the United States are fundamentally different, determining their unique characteristics of models. China aims to diffuse the AI application into various industries and fields, such as humanoid robots and vehicles (Chan et al., 2025), while the United States desires to maximize the compute capability and design the most powerful model (Chan, 2026). In other words, the former is adopting AI technology horizontally, while the latter is striving to go deeper in the vertical sense, achieving frontier models and AGI (Chan, 2026).

Marking its significance, embodied AI, AI embedded in physical machines such as robots and vehicles that can sense and act in the real world, is prioritized in China's 15th Five-Year Plan (2025). According to a study by IFR in 2025, China has already produced around 2 million industrial robots, which is 4.5 times the number in Japan, taking up 54% of global new installs. Financially, it receives a total amount of \$138 Billion national VC guidance fund to support its development (Chan et al., 2025). Another industrial reason underlies the developed supply chain: China has a mature robotics manufacturing procedure, which makes scaling up the production easier.

As mentioned before, chips produced by NVIDIA and AMD are compute-intensive, providing sufficient compute power to meet the intensive requirement of frontier models. In the past few years, the United States purchased a large amount of such hardware to meet the rising domestic demand, buying as many as 485,000 Nvidia Hopper GPUs from Nvidia in 2024, as many as Microsoft alone did ("Microsoft bought nearly 500K", 2024). This allowed research labs and tech companies like Anthropic, Google Deepmind, and OpenAI, who are the main contributors to the frontier models, to develop compute-intensive models without worrying about the deficiency of chips. Artificial General Intelligence (AGI), an AI system "with human-level ability to learn and apply knowledge across a wide range of tasks and domains" (Stanford HAI, n.d., para. 1), thus believed to be achieved in the coming future. It will empower humanity with an unimaginable amount of ability, promoting human ingenuity and creativity, noted in Open AI's article Planning for AGI and beyond (2023).

On the other hand, as the United States has imposed an AI chips import ban on China, it successfully limited the hardware supply for testing advanced models, forcing the eastern rival to rely on distillation – “uses the outputs of a powerful AI system to train a smaller model that can

perform some of the same tasks with fewer computing resources” (Baptista, 2026, para. 3) – optimize the model efficiency, and reduce the operational compute. For instance, compared to Meta's Llama 3.1 405B consuming around 30.8 million GPU-hours of compute, DeepSeek-V3 only requires about 9% of compute to produce nearly 100% of the U.S. counterpart's efficacy (DeepSeek-AI, 2024). This feature has also allowed Chinese tech companies to charge a relatively low price for their customers, creating another competitive advantage. For example, a San Diego developer who switched from Claude to DeepSeek found that an hour-long coding session costing about \$10 on Claude cost less than 50 cents on DeepSeek, remarking simply, "You don't need God to write your email" (Rest of World, 2026, para. 7).

Another discrepancy lies in the transparency of weights, which is the concept of how far a model's trained parameters are made publicly available for anyone to download, inspect, and reuse. Although the US AI action Plan published by the White House (2025) endorses open source, most frontier models, including ChatGPT, Claude, and Gemini, are proprietary and closed to avoid their rivals taking advantage of this critical information and using it against them. A further rationale is safety-related: open weights make it "very difficult to apply guardrails to them or monitor their usage," and once released they cannot be withdrawn (Amodei, 2026, para. 5). However, the majority of Chinese counterparts, supported by governments, disclosed their API and weights to allow international access. This has attracted a significant number of users to adopt Chinese open-weight frontier models, largely leading to the drastic increase from 1.2% to 30% of global AI usages within 12 months (Axis Intelligence, 2026). Although Chinese companies are not directly financially benefiting from this action, they have gained renowned global reputation and influence over revenue (Chan, 2026).

Advancement and innovations in patents in both countries have been conspicuous. In terms of GenAI patents registered from 2014 to 2023, China has an overwhelming number at 38210, almost six times more than the 6276 for the United States (World Intellectual Property Organization [WIPO], 2024). This might result from the large AI-based professionals' population in China due to demographic reasons. Granted, the practicality and usefulness of the U.S. patents appear higher than that of China. Each US patent will be cited at an average time of 13.18, while its Chinese counterpart will only be cited 1.9 times (WIPO, 2024). However, a reconciling reason for this phenomenon is that most Chinese cooperatives only apply for patents domestically, lacking international exposure. In terms of publication in 2023, China constitutes 22.6% of the global AI reference citations, twice as much as the US (Stanford HAI, 2025). Although the top 100 is still composed mainly of studies from the US, there's a growing trend that the most cited papers are China and have increased from 33 in 2021 to 41 in 2024 (Stanford HAI, 2026). Comprehensively speaking, both entities have been contributing to the research of AI technology, and the US is currently taking the lead at a slight advantage.

Infrastructure and Data

Infrastructure is where two NIS diverge the most physically, and the discrepancy lies within the organizing logic instead of resource quantity. The United States tends to treat computers as a commodity and requires enough power to feed it, while China considers it as a public utility and allocates it to the place where power is adequate. This logic is the rationale for various actions conducted by governments, tech companies, and other actors.

The supply of electric energy is a critical factor for AI development and considered during decision making. Power demand has been exploding in both regions as data centers are consuming tremendous amounts of energy. On one hand, although the energy added

543 Gigawatt in 2024 alone (Institute of Electrical and Electronics Engineers Communications Society [IEEE ComSoc], 2026), China has been expanding its national power grid over the past few decades, initiating even before 2010. Thus, it sometimes faces surplus in energy and overcapacity, and the price of power does not fluctuate drastically due to the stable supply. Institute of Electrical and Electronics Engineers (IEEE) estimates that 400-Gigawatt spare capacity will exist by 2030, which is around three times as much as the total global data-center energy demand.

On the other hand, the United States is currently building up power after realizing its importance before long. Even so, some regions are facing severe shortages: the price of electricity in PJM, where the 13-state grid operator containing the world's largest data-center cluster in Northern Virginia, increased more than 833% from 2025-2026 compared to the previous year (American Action Forum, 2026). Additionally, Morgan Stanley (2025) projects that in 2028, the United States will experience a 44-Gigawatt energy shortfall. These shortfalls leave firms unwilling to wait on a strained grid. Compounding the problem, the slow bureaucratic process made companies seek independent power generation. Connecting a power plant or AI data center into the national grid might take up to seven years. To circumvent the complicated procedures, most data centers decide to generate power on-site by adopting Nuclear Power Purchase Agreement, purchasing carbon-free electricity directly from a nuclear plant, or using natural-gas turbines (Morgan Stanley, 2025).

However, abundant, cheap power in China has failed to be converted into efficient usage in data centers, heavily diluted by large-scale idleness, undermining the "China's energy is a crushing advantage" narrative. In the grand project of East Data West Computing, relocating data processing and storage tasks from China's densely populated eastern coastal areas to

resource-rich western inland provinces, 633 large and hyperscale data centers were constructed with 268 EFLOPS of compute (Australian Strategic Policy Institute [ASPI], 2026). Nevertheless, the average utilization rate among data centers is only ranging from 20-30% (Yahoo Finance, 2025), indicative of wide idleness and misallocation of resources, while that of western counterparts is assumed to be over 70% (Goldman Sachs, 2026). Thus, energy surpluses are not quite equivalent to advantage, as resources might be wasted and costs imposed on local governments and State-owned Enterprises (SOEs).

Data has proved to be an indispensable asset in the era of AI, since it effectively trains frontier models to spontaneously analyze and learn the underlying patterns and rules on its own, then apply them into real-world scenarios. Undoubtedly, the two countries have distinctive data regimes: the U.S. AI models depend on private, commercial licensing, high-quality public text corpora, yet might be facing rapid exhaustion from 2026-2032 (Epoch AI, 2024), while China generates abundant data from physical deployment, surveillance networks, industrial Internet of Things (IoT), and embodied settings. Secondly, the understanding of what data really is also diverges between two NIS. From the western perspective, data is a private commercial, firm-held asset without any federal designation. A perfect explanatory example is that Anthropic purchased millions of physical books and destructively scanned them into a digital corpus to train its frontier Claude models. In court rulings, this act is a form of legal “format shifting” and Fair Use according to the US copyright law (Bartz v. Anthropic PBC, 2025). The inverse is true in China, while in 2020 the State Council formally designated data as the "fifth factor of production" alongside land, labor, capital, and technology, indicating that the government will be able to directly manage it (DigiChina, 2022). A National Data Administration was inaugurated in October 2023 and required firms to register data assets on their balance sheet (Lexology, 2024).

Speaking of regulatory laws, China has centralized and comprehensive rules – Data Security Law, amended Cybersecurity Law, and Personal Information Protection Law forming strict regimes on paper (Chambers and Partners, 2026) – while the United States has mandatory yet sectoral laws standing in different industries (health, finance, kids) and by state (ICLG, 2026).

The diverging in infrastructure and data can lead to one institutional fact: the United States’ decision is mainly driven by private companies and the market itself, while the Chinese government has centralized power and dispatch resources to where it is needed. (National Foundation for American Policy [NFAP], 2025)

Allocation and Flow in Human Capital

The World Economic Forum’s Future of Jobs Report 2025 ranks AI specialists as the single fastest-growing occupation, projecting around 40% annual growth from now until 2030. Top-notch STEM researchers and developers are critical to the study of frontier models. In this case, the United States and China have exhibited radically different situations: China is the world's largest supplier of AI talent, while the United States is the largest employer.

A study by the Center for Security and Emerging Technology (CSET, 2021) estimated that more than 77,000 Chinese STEM/AI Ph.D. s have graduated from educational institutes in the year of 2025, while only around 40,000 in its western counterpart. Excluding international students in the United States, China “out-produces” the United States about more than 3 to 1. Moreover, 47% of the world’s top-tier (approximately 20%) AI researchers in 2022 are from China, compared to 29% in 2019 (MacroPolo, 2024). On the other hand, the United States contributed 18-29% of top-tier researchers. These statistics are valid to corroborate the claim that mass production of AI elites is happening domestically in the eastern entity rather than in the western. One reason for this phenomenon is the large demographic base, the Chinese taking up

roughly 18% of the world's total population (United Nations, 2024), so even a normal share choosing STEM will yield huge absolute numbers. Secondly, the Chinese educational system since 1999 has been deliberately built to serve the STEM pipeline, by methods like mass university enrollment expansion, and a sustained scale-up of graduate and doctoral programs. Lastly, hundreds of universities and talents programs were directed to open AI majors and expand enrollment after the 2017 national AI plan (State Council, 2017), as talent production becomes a policy target instead of just a market favor.

Interestingly, although the United States does not train human capital by itself, AI professionals from various countries, especially for India and China, are still concentrating in its region. According to the data from MacroPolo's Global AI Talent Tracker (2024), it employs 57% of the top 2% of global AI talent, remarkably contributing to the development of frontier models. More than half the U.S. AI workforce is foreign-born, and in 2024, 61% of U.S. CS PhDs earned their bachelor's degree (around 23% is from China) (NFAP, 2023). Among the Chinese AI PhD students who have worked in the United States, around 80% of them historically stay, and less than 20% of those who leave go back to China (CSET, 2023). Reasons for U.S. attractions are conspicuous and encouraging. 60% of global top AI universities and the leading labs like OpenAI are located here (MacroPolo, 2024). Moreover, the United States offers the highest AI salaries, equity, a mature startup and VC ecosystem, and an open, meritocratic research culture. Frontier labs compete aggressively on pay and equity, the United States draws the majority of global private AI investment—far exceeding China (Stanford HAI, 2025)—and its open academic culture lets researchers publish and collaborate at the frontier. Immigration pathways are one important reason as well, since the route of an F-1 student visa, Optional

Practical Training (OPT), an H-1B work visa, and eventually a green card allows talents to yield long-term work authorization and be able to stay.

Although the U.S. talent magnet was strong and effective historically, it started to be thwarted in recent years. In 2022, 42% of top tier researchers worked abroad, which goes down 13 points from 2019 (MacroPolo, 2024). More scholars start to prefer staying in home countries, indicative of falling overall mobility and rate of influx. In August 2025, international student arrivals declined 19% year-on-year, where Chinese students held the number at around 12% (NFAP, 2025).

The U.S. government has conducted several actions to prohibit certain talents, especially targeted towards Chinese, from studying and working in the nation to ensure national security and restrict the development of the Chinese AI industry. In 2018, the United States began to act more strictly on Visa, cutting visa validity for Chinese graduate students in sensitive fields (robotics, aviation, advanced manufacturing, etc.) from 5 years to 1 year (Zengerle & Spetalnick, 2018). In May 2020, the White House barred F-1/J-1 entry to Chinese graduate students and researchers associated with China's "military-civil fusion" strategy (Proclamation No. 10043, 2020), which affected an estimated number of 3000-5000 people per year (Niskanen Center, 2025). In September 2025, a proclamation imposes a \$100,000 fee on most new H-1B petitions and restricts entry of certain holders, raising the barrier for some foreign researchers to stay in the United States (The White House, 2025a). Though struck down by a judge, it still remains in effect pending appeal (Manifest Law, 2026).

Economic Capital and Financing

In order to encourage and empower AI development more practically, financial investments from private firms and governmental subsidies will be required. AI startups took

around 46% of global VC in Q3 2025 (\$97B) alone, roughly half of all venture funding for the year (Crunchbase, 2026). Thus, it is important to allocate capital in the right spot and convert it into capability more efficiently.

In terms of private-investment metric, the United States takes the lead in magnitudes, and the gap is projected to even widen in future years. In 2025, the U.S. private AI investment reached \$285.9B, 23 times greater than China's \$12.4 Billion investment, more than that in the United States in the previous year (Stanford HAI, 2026). Within this, more than half of the total amount (\$163.6 Billion) is targeted towards Generative AI, which can create original content in response to a user's prompt or request (Stanford HAI, 2026). This is primarily because the United States is primarily VC-dominated and has formed a mature and deep ecosystem, fueling the startups, frontier labs and companies. In fact, VC has provided 79% of global AI funding for U.S. startups in 2025 (Second Talent, 2026), compared to the 48% in 2023, while Asia's share fell from 30% to 13% (World Intellectual Property Organization [WIPO], 2024). California alone took up \$218 billion last year, more than 75% of the total amount (Stanford HAI, 2026). This is mainly because a large share of AI related tech companies and research labs are located in Silicon Valley and are home to flagship frontier labs such as OpenAI, Anthropic, and xAI, thus requiring a tremendous amount of economic input to satisfy their needs. The Bay Area alone captured 60% of global AI funding in 2025 (The AI Economy, 2026).

Beyond startups, the private buildout is led by mega-caps spending off their own balance sheets, especially in the western community. The Big Four – Amazon, Microsoft, Alphabet, and Meta – have collected \$725 Billion in 2026, up from \$410 Billion last year (James, 2026). Stargate, a flagship project presented by OpenAI, SoftBank, and Oracle dedicated to making further progress in AI infrastructure, has raised around \$500 Billion since January 2025 (Open

AI, 2025). Meanwhile, in China, most companies do not have the equivalent scale of fundraising: Alibaba alone has made around \$53 Billion over three years (Alibaba Group, 2025).

On the other hand, the Chinese government does not stay idle for investment. Instead of private funding, state-directed capital is more crucial for tech firms, since the government directs the general operations and centralizes financial power. It is key to sustaining chips and AI bets despite no significant profit within years. Government guidance funds have dispatched \$184 Billion to AI firms from 2000-2023 (Digital Strategy, 2025), and in 2025 alone frontier organizations received a total amount of \$138 Billion state VC fund (Chan et al., 2025). The country has established a specialized project for developing advanced chips which amounted to \$47.5 Billion last year (Stanford HAI, 2025), attempting to mitigate the hardware disadvantage brought by the export control in relation to its western rival. However, the entire distribution logic might be prone to misallocation of resources since the decision-making directly and solely depends on the will of the Chinese Communist Party (CCP), compared to the demand-supply driven strategy in the United States. Although the U.S. government is also starting to intervene financially, deploying the state equity in more than 30 companies since 2025 (Yahoo Finance, 2026) – for instance, converting around \$8.9 Billion in CHIPS Act into a 9.9% equity stake in Intel (Manufacturing Dive, 2025) – the transparency of funding and its degree fails to catch up with China.

Policy, Regulation, and Governance

Policy is where the institutional structure behind every prior dimension becomes explicit, and both countries have been implementing regulations to follow up the rapid development of technology and setting incentives to encourage the study of frontier models and industry

adoption. The United States and China do not share the same ideology or political standards, thus demonstrating wide divergences in framing those action plans.

On the promotional side, both countries actively steer AI in their industrial strategies. However, China does this via explicit central operation, while the United States goes through a market-mediated policy. In 2017, the New Generation Artificial Intelligence Development Plan in China was released by the state council, acting as the first official document explicitly stating the direction of future AI development, also setting the founding ambition of becoming the innovation center by 2030 (State Council, 2017). The 15th Five-Year Plan ranging from 2026-2030 put more emphasis on concrete deployment of models into industry and the concept of embodied-AI rather than general encouragement (Central Committee of the Communist Party of China, 2025). Firms and SOEs follow these guidelines proactively, distributing more focus and resources on the AI industry to fulfill the objective presented in those central plans. The local government has made more detailed initiatives and engaged in associated programs such as "East Data West Computing" and talent recruiting. On the contrary to the top-down structure in China, the white house had few equivalent mandatory central AI plans, but rather fragmented actions based on the market performances. Private labs and capital set their direction of AI development while the government responds by supporting and clearing obstacles, representative of the bottom-up approach. EO 14179 "Removing Barriers to American Leadership in AI" as a great example, ordered agencies to strip away regulations seen as barriers as most companies are facing bureaucratic procedures during the infrastructure construction and supervision phase (The White House, 2025a). Published in July 2025, America's AI Action Plan has provided 90 federal actions, yet functioning as a directional signal rather than binding laws like its counterpart in China (The White House, 2025b).

In terms of regulatory architecture, the systems diverge sharply because of their distinctive political conditions and institutional structure. On one hand, China is constructing an enforced, expanding regime by making central laws, requiring states and firms to focus on the control of AI and data security. This can be illustrated through three foundational statutes: the Cybersecurity Law (2017, amended in 2026 to include more AI related policies), the Data Security Law (2021), and the Personal Information Protection Law (2021). All of them are enforced by the Cyberspace Administration of China (CAC) (Deep Lex, 2026). China is also the first country to impose binding rules on GenAI by announcing the Interim Measures for the Management of GenAI Services in 2023 (Deep Lex, 2026). Moreover, algorithm filing is a precondition before officially launching in the market, and 796 generative AI services in total have been registered by February 2026 (Deep Lex, 2026). This can effectively ensure the government evaluates the safety and supervises the models in advance, preventing risky situations from occurring after the release. Most recently, to cope with deep-fake and the spread of misinformation, the government also requires all AI-generated content to carry explicit and implicit labels, with fines reaching up to 15 million RMB or 5% of annual turnover, backed by the mandatory national standard GB 45438-2025 (Lexology, 2026). On the other hand, in January 2025, EO 14110 was rescinded (Baker Botts, 2026), which is an order that had imposed binding reporting obligations on the developers of the most capable frontier models under the Defense Production Act. Safety-test sharing and chief AI officer mandates were no longer required. The government will only act when the model has committed serious dangers. The market-access control is also withdrawn, which means unlike China's algorithm filing, no federal licensing is needed for frontier AI models in the United States (Baker Botts, 2026). These similar policies have exposed that the White House will comply with the baseline of "innovation-first"

and reduce as much friction as possible on the road of developing AI, where the power of state is manifest. However, such freedom and lack of regulation might cause backlash and bring data leakage to the general public, which the government tends to neglect. "Deregulation threatens public expectations and safety" (Electronic Privacy Information Center [EPIC], 2025, para. 2).

To conclude, both nations have been implementing laws promoting the development of AI. However, that in the United States is more agile but contested among analysts, since it tends to be more sectoral, market-led, and radically encourages innovation by clearing out safety rules. In China, the policies are centralized, mandatory, and coherent; thus, they might appear to be more manipulative and conservative, yet the cybersecurity is better accentuated compared to the western nation.

Three Types of Interactions between Actors

As mentioned before in the NIS framework, Lundvall's core claim is that the communication between organizations and how each influences the other is particularly worth studying since it induces the emergence of innovation (Lundvall, 1992). This paper has categorized the actors into four main types: state, firms, universities, and finance (discussed already in "Economic Capital and Financing"). Similar to the previous section, NIS in the United States shows a decentralized market network, while China exhibits a state-orchestrated web.

The connection between state and firms directly determines the shape and foundation of the entire network. The White House and AI-associated Administrations sit beside firms as a facilitator, giving them enough freedom to explore on their own and respond to the market's demand. The policy section has illustrated that instead of setting a binding direction, the federal government strives to minimize friction on the way of development, including safety tests and market access control. In turn, the firms have sold their product to the government to better assist

them in decision-making and promoting productivity. Announcing that Claude, Gemini, and ChatGPT had been applied governmentwide, Acting General Services Administration administrator Michael Rigas said the state was "leveraging the private sector's innovation to transform every facet of government operations", aligning with the AI Action Plan (FedScoop, 2025, para. 4). In July 2025, the DoD's Chief Digital and AI Office awarded contracts of up to \$200 million each to four major AI companies to accelerate the department's adoption of advanced AI capabilities for national security (Nextgov/FCW, 2025). This reciprocity has also been apparent in China as well, yet the state council is more control-oriented and centralizes its power to enforce firms to follow the general developing agenda. The Five-Year-Plan has induced cascades of following actions in companies, and the latter will be eligible to receive national funding (Chan et al., 2025). Beyond funding, the state also embeds itself directly into corporate governance through a strategy named "golden shares". Since 2021, Cyberspace Administration of China has acquired roughly 1% "special management" stakes in tech companies such as ByteDance, Alibaba, and Tencent (CNN Business, 2023). Though it seems to be negligible, these stakes actually mean the state can directly intervene in the actions made by the board of the firms. Military-civil fusion is also an important agenda in China, fuses the commercial and national defense to produce more innovative products, making the state a co-developer rather than a customer. For instance, iFlytek, a national champion in speech recognition, supplies the People's Liberation Army with AI-enabled translation and decision-support systems (Foreign Policy, 2025).

The communication between research institutes and the industry is also a critical aspect since it fosters the development via training professional scholars. Both systems have now exhibited the same trend that industry has replaced academia as the site of frontier model

production, since it can respond to the market preference and attract funding more directly: Stanford HAI's 2026 AI Index stated that the global AI industry produced over 90% of notable frontier models in 2025, a rise from only around 60% in 2023 (Stanford HAI, 2026). Some universities have established labs with companies to foster research and commercial interest: MIT-IBM Watson AI Lab established in 2017 (MIT-IBM Watson AI Lab, 2017) and Stanford Human-Centered AI supported by Google, Accenture, Toyota etc. (Stanford HAI, n.d.). When a research lab has yielded groundbreaking progress or designed a product fulfilling market demand, VC will provide funds for them to expand. Some of them will be bought out by large tech firms. This indicates that most cooperation between research institutes and companies is designed for commercial purposes. However, those collaborations in China are tightly aligned with the national strategic agenda like the Five-Year-Plans. For example, in 2020 Huawei partnered with the Ministry of Education to launch the "Intelligent Base" program, embedding its Ascend AI curriculum into leading universities such as Tsinghua and Peking University to break down the barriers between academia and industry and to support a fully domestic computing stack rather than relying on imported components; its Ascend ecosystem now works with more than 55 universities (Huawei, 2025). Tsinghua also leads national key research and development programs on optimizing computing systems for domestic AI accelerator chips, and its computer-science teams have adapted models such as DeepSeek to run efficiently on Huawei's Ascend hardware (Tsinghua University, 2024; Department of Computer Science and Technology, Tsinghua University, 2025).

Lastly, the competition and cooperation between firms have also driven the innovation and diffusion of AI technology. The U.S. frontier race based on parameter scale between firms like OpenAI, Anthropic, Google, xAI is compute and financial-intensive, often in the form of

one releasing powerful models and providing additional usage. For example, in mid-2026 OpenAI signaled it was ready to sharply cut the price of its GPT-5.6 models to win users from Anthropic, with chief executive Sam Altman noting that the model was already half the price of Anthropic's Claude Fable 5 and offering to lower it further (South China Morning Post, 2026a). In China, companies are in the “Battle of Thousand Models.” With hardware restraints resulting from U.S. export bans towards chips and important machinery, the competition has shifted from hardware brute force to algorithmic and cost efficiency. Most Chinese models are renowned for their low compute cost yet providing reliable results, like Kimi / Deepseek; for instance, DeepSeek’s V4 model matched leading Western systems while advertising drastically reduced compute and memory costs, pricing its output at a small fraction of its U.S. rivals (Fortune, 2026). All competitions will eventually benefit the consumers with more credits to use, lower price, and higher efficiency. As one senior industry analyst observed, the sector has entered "the era of efficiency," in which the provider that delivers the most capability at the lowest cost wins — a dynamic that steadily lowers prices for users (The AI Chronicle, 2026).

Collaborations between firms are often complementary: Microsoft supplies Azure computers with infrastructure, while OpenAI provides exclusive models in turn (Microsoft, 2023); similarly, AWS and Google offer multi-billion-dollar computer partnerships providing infrastructure in exchange for Anthropic's model deployment access (Amazon, 2024). In China, DeepSeek works directly with Huawei to optimize its Mixture-of-Experts (MoE) reasoning models (their V4 model) for Huawei's Ascend NPU architecture (the Ascend 950 series) (South China Morning Post, 2026b).

Emerging Economic and Geopolitical Consequences

The divergent capital allocation mechanisms of the U.S. and Chinese NIS have created distinct macroeconomic growth patterns and economic vulnerability. In the United States, the economy heavily depends on VC investment, as venture-backed firms account for roughly three-quarters of the market value of U.S. public companies founded since the late 1970s (Gornall & Strebulaev, 2021). Moreover, construction of AI infrastructure has contributed to half of GDP growth in the past year, with 92% for the first half in 2025 (Soto et al., 2026; Oliver Wyman, 2026). The top 10 stocks in the S&P 500 account for 35% of the index (Intellectia, 2026). This suggests that the U.S. stock market was heavily influenced by the AI movement led by Microsoft, Apple, Nvidia, Google, Amazon, and such tech companies. These indications make it prone to the named "Single Point of Failure" by Apollo, stating if the downstream application is far below expectations, a stock market pullback will directly trigger a macroeconomic recession (Apollo Global Management, 2026). However, in China, the governmental guidance funds summing up at \$912 Billion (Axis Intelligence, 2026) have distributed risks at early investment stage to public balance sheets like SOEs and local governments (Stanford Center on China's Economy and Institutions [Stanford SCCEI], 2024), attempting to minimize the effect stock market might bring to ordinary citizens (Hove, 2026).

The physical expansion of AI data centers and national grid creates externalities associated with costs. Utilities in the United States have incorporated the cost of upgrading the electric grid into a rate base. Goldman has predicted that the explosion in the amount of energy consumed by data centers will lead to a national average increase in price of around 6% (CNBC, 2026). In fact, the residential electric price in Northern Virginia has already increased 260% within the past five years (The Motley Fool, 2026). While large tech firms and electric companies have made considerable profits, the cost is imposed on innocent citizens. On the other

hand, the Chinese do not experience an influx in electric price since it is operated by the central government rather than the market. However, some data centers constructed in remote regions have a utilization rate as low as 20%-30% (Chen, 2025), representing serious misallocation of public resources, as discussed in the previous sections (Chan et al., 2025).

The two NIS modes also drive different industrial influences. In the western context, the industrialization of AI will accelerate white-collar automation in knowledge sectors. Although stated as "complementary but not supplemental" in AI Action Plan (2025), most preliminary and intermediate workforces face the possibility of being replaced by technology. For example, in September 2025 Salesforce eliminated roughly 4,000 customer-support roles after its chief executive stated that AI could already handle about half of the company's customer interactions (CNBC, 2025). However, the federal government does not provide any official re-employment assessment or transition support. In contrast, AI is applied in manufacturing and embodied AI according to the CCP's 15th Five-Year-Plan (2026). It shall be widely promoted via MoE at relatively low cost (DeepSeek-AI, 2024). Also, the state has encouraged the firms to engage in training their employees to adapt to new skills and provide help during the transitional phase. For instance, China has launched a nationwide skills-upgrading campaign running through 2027 that provides subsidized training for more than 30 million workers, together with retraining subsidies and re-employment incentives for those displaced by technological change (State Council Information Office, 2026).

On the international stage, the two different NIS have attracted countries in various continents to form alliances with them and adopt their frontier models. In America's AI Action Plan (The White House, 2025b), it has used Foreign Direct Product Rule and Secondary tariffs to force allies to align with the restrictions, constructing an exclusive, minilateral (involving a small

group of countries working together to address a particular issue) U.S. AI alliance (e.g. UK, Japan, Singapore, Australia). In China, the Ministry of Foreign Affairs released the Global AI Governance Action Plan (Ministry of Foreign Affairs, 2025), appealing for the United Nations to serve as the primary channel of supervision and knowledge diffusion. The government wants to establish a relationship with countries in the "Global South" on the opposite side of the U.S. alliances, including Indonesia, Kenya, Malaysia etc.

Discussion

Answer to the Research Question

In response to the central research question, while China and the United States share common pressures such as GPU hardware bottlenecks, soaring electricity demand, and talent competition, their developmental pathways are mostly shaped by opposing National Innovation System (NIS) frameworks. The United States is mostly market-driven and decentralized, with its federal government facilitating the radical moves of tech firms. On the other hand, China's framework is state-coordinated and power-centralized. The state council directs the general agenda and requires local governments, SOEs, and private companies to proactively follow it.

In terms of technical research, the United States pursues vertical, frontier, compute-intensive capability, while China focuses more on cost-effectiveness, the diffusion of technology into industries, and embodied AI. In financing, the former entity relies heavily on private VC and Big Four's CapEx, while Chinese governmental guidance funds play a critical role in the developmental stage. For infrastructure, the United States might potentially experience power shortage and treat the computing power as a market commodity, while overcapacity and underutilization is occurring in China, and the government is considering computing power as a factor of production in future era. The attitudes toward policymaking also

vary: the United States acts after accidents happen and support less strict supervision, but the Chinese central state takes control in advance and designs strict market-access regulations.

Ultimately, these institutional mechanisms result in distinct economic and political consequences: the U.S. macroeconomic strong reliance on AI might lead to higher stock market risks when expectations are not met. Increases in energy prices due to the tremendous consumption by data centers is another problem. In China, the misallocation of resources and fiscal burden imposed on local governments remain to be resolved. Both countries are diffusing their models and impact internationally as a future geopolitical strategy.

Social Circumstances and Distributional Effects

The fundamental reasons why two nations have distinct NIS are deeply rooted in demographic conditions, cultural expectations of governance, ideological differences, and national development agendas. China's political power is solely concentrated at the CCP. Thus, they use a more control-oriented mechanism to secure their governance, cybersecurity, and livelihood of its people. Facing an ageing population, drop in workforces, yet still heavy reliance on the manufacturing industry, it will focus more on adopting AI automation into such processes to address the emerging issues. On the contrary, the United States is dominated by free market, individualism, and capitalistic narratives instead of socialism. Thus, commercial outcomes and frontier breakthroughs are core in AI development, and firms are actively responding to market preferences, preventing them from being controlled by the federal government.

Each innovation system inherently embeds the priorities of powerful actors while marginalizing other risks. In China, the political governance, advantages in manufacturing, and national securities are the top priorities, while local governments and resource-deficient inland states are less concerned. In western societies, the economic profits of VC, main shareholders,

and STEM elites are benefiting from this mechanism, while ordinary residents and traditional white-collar workers are more neglected.

Implications

This paper has provided practical insights for policymakers, researchers and firms by analyzing the fundamental discrepancies between the National Innovation Systems of the United States and China. For U.S. policymakers, they should establish frameworks to distribute the cost of electricity more evenly rather than imposing most of them on citizens, and the data security and control of AI shall be better safeguarded. For the Chinese counterparts, they need to promote the utilization rate of certain data centers and ponder more thoroughly while allocating resources, by listening to more professional analysts and observations of local conditions.

For firms and industry leaders in the west, they should consider promoting algorithm efficiency instead of brute-compute-force-scaling, to better address the power shortage and “single point of failure” in the stock market. Chinese firms and companies shall continue to provide open-source models for the expansion of the global developer ecosystem and comply with the algorithm filing and related safety control regulations.

For workers and citizens, both countries need to build up a system of evaluating and mitigating the impact by implementing AI into traditional fields as soon as possible to secure the livelihood of preliminary and intermediate workforces. Residents need to realize the possibilities and consequences of being replaced by AI technology and train themselves with more unique skills.

Conclusion and Limitations

This paper aims to comprehensively evaluate how similarly and differently China, and the United States organizes research, capital, infrastructure, regulation, and labor for Artificial

Intelligence, by applying the National Innovation System Framework instead of the oversimplified and fragmental “technological race” narratives. Analysis towards different dimensions have revealed that distinct institutional logics (centralized or market-driven, socialism or capitalism) – formed by a nation’s history, comparative advantages, economy and politics – can lead to entirely different objectives and outcomes.

However, several methodological limitations must be acknowledged. Firstly, the algorithm and policy paradigms shift rapidly, so this study can only offer insights into the 2020-2026 period. Secondly, the standardization of two countries’ paper citations, investments, computing powers, and related indices might include systematic errors. Lastly, some private research labs and SOEs do not fully disclose their data including model training costs, actual utilization rate. Some can even be manipulated for certain purposes.

In conclusion, the AI rivalry between the United States and China demonstrates that the future of economy, state capacity, and global geopolitical order will not be solely determined by artificial intelligence technology in isolation, but also by how nations organize, govern, and institutionalize innovation itself.

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Signature: Ziheng Wang Date: 08.18.2026

